TalentTrace AI - Database Design

PostgreSQL + pgvector | Core Business Data | Evidence Traceability

PLANNED TARGET SCHEMA - implementation not implied

EVIDENCE PATH

JobProfile -> CandidateApplication -> ScreeningResult -> InterviewSession -> Evaluation -> ConsolidatedAssessment

1 - JOB & CANDIDATE FOUNDATION

User

PK id
email UNIQUE
password_hash
role
is_active

Recruiter/interviewer only • no candidate account

JobProfile

PK id
title
version
status
FK created_by -> User.id
scoring_weights JSONB

JobRequirement

PK id
FK job_profile_id -> JobProfile.id
description
category
is_mandatory
importance

Candidate

PK id
name
email
phone
created_at

Identity excluded from scoring

CVDocument

PK id
FK candidate_id -> Candidate.id
storage_path
original_filename
mime_type
uploaded_at

CV file in private volume

JobPublication

PK id
FK job_profile_id -> JobProfile.id
provider
external_post_id
status
published_at

2 - CORE APPLICATION & SCREENING

CandidateApplication

PK id
FK candidate_id -> Candidate.id
FK job_profile_id -> JobProfile.id
FK cv_document_id -> CVDocument.id
FK screening_batch_id -> ScreeningBatch.id NULL
status
applied_at

CORE BUSINESS RECORD • Candidate + specific Job Profile

ScreeningBatch

PK id
FK job_profile_id -> JobProfile.id
status
total_candidates
completed_count
failed_count

ScreeningResult

PK id
FK application_id -> CandidateApplication
hard_requirements_passed
technical_score | experience_score
qualification_score | ats_match_score
aggregate_score
status
justification | model_metadata JSONB

RequirementMatch

PK id
FK screening_result_id -> ScreeningResult.id
FK requirement_id -> JobRequirement.id
status
evidence
evidence_location
justification

Met • Partially Met • Unmet

3 - INTERVIEW & EVALUATION

InterviewSession

PK id
FK application_id -> CandidateApplication.id
scheduled_start | timezone
access_token_hash
token_expires_at
meeting_provider
status

InterviewEvaluation

PK id
FK interview_session_id -> InterviewSession.id
status
model_metadata JSONB
created_at

TranscriptTurn

PK id
FK interview_session_id -> InterviewSession.id
sequence_number
speaker
started_at | ended_at
text

Final STT turns only • no raw AV

QAPair

PK id
FK interview_evaluation_id -> InterviewEvaluation.id
question_text
answer_text
question_turn_refs / answer_turn_refs JSONB
classification

RequirementAssessment

PK id
FK interview_evaluation_id -> InterviewEvaluation.id
FK requirement_id -> JobRequirement.id
coverage_status
relevance_score | accuracy_score
completeness_score | clarity_score
evidence_gaps JSONB

ConsolidatedAssessment

PK id
FK application_id -> CandidateApplication.id
overall_score
overall_suitability
strengths / weaknesses JSONB
evidence_gaps JSONB
recommendation_summary

Human final decision

4 - KNOWLEDGE & WORKFLOW PERSISTENCE

KnowledgeDocument

PK id
knowledge_type
title
source

KnowledgeChunk

PK id
FK document_id -> KnowledgeDocument.id
chunk_text
metadata JSONB
embedding VECTOR

RAG Boundary

company_policy
technical_reference
CVs / jobs / transcripts are business data

LangGraph Checkpoint Persistence

Workflow resume / recovery state

Private candidate_uploads Volume

PDF / DOCX CV files
DB stores references only

Planned schema • Evidence traceability • Coordinated database and private-file cleanup